

Martin Nunction

Martin Nunction (a pun for Notation/Function), Is a large number notation created by Martin Wanralfs or UpBeatSquirrel925 in Fandom wiki.

A. Definitions

1. Martin Aunction (MA)

Martin Aunction is the very first function in Martin nunction, written as:

$$A(a,b,c)$$

Let a be any non-zero finite positive integer, a is the argument, a is the first entry.

Let b be any non-zero finite positive integer, b is the amount of (c-th) hyperoperation, b is the second entry.

Let c be any non-zero finite positive integer, c is the hyperoperation level, c is the third entry.

General Martin Aunction expansion:

$$A(a,b,c) = a \uparrow^c b \quad a = a \uparrow^{c-1} b \quad (a \uparrow^{c-1} b) \uparrow^{c-1} b \dots \#.$$

Right associatively, using a value as the amount of iteration.

(Note: octothorpe symbol means unchanged remainder of an array and hyperoperation.)

$$A(a,b,c) = a \uparrow^{(c\text{-th level hyperoperation})} b \quad a.$$

General Martin Aunction leveling down:

$$A(a,1,c) = a \uparrow^{(c-1)} a \quad a, \text{ or}$$

$$A(c)^1 n = a \uparrow^{(c-1)} n.$$

Where b = 1, b value replaced by right operand a, while c is subtracted by 1, moving to (c-1)-th hyperoperation.

Examples:

$$A(1,b,c) = 1 \uparrow^c b \quad 1 = 1$$

$$A(1,1,1) = 1 \uparrow^1 1 = 1^1 = 1$$

$$A(1,n,n) = 1 \uparrow^n n \quad 1 = 1$$

Where c = 1 is the Knuth's up arrow hyperoperation ($\uparrow^c$), a = 1, left a is the argument while right a is hyperoperation expansion operand, and b = 1 is the amount of current level hyperoperation, a becomes the result ignoring the second and third entry iff a = 1.

$$A(2,2,2) = 2 (2)^2 2 = 2 (2)^{12} = 2 \uparrow^2 2 = 2 \uparrow 2 = 2^2 = 4.$$

$$A(2,b,c) = 2 (c)^b 2 = 2 (c)^{b-1} 2 = 2 (c)^{b-2} 2 \dots$$

Operation will terminate to 4 such that $a = 2$.

$$A(3,2,1) = 3 \uparrow^2 3 = 3 \uparrow (3 \uparrow 3) = 3 \uparrow 27 = 7,625,597,484,987.$$

2.Martin Array Aunction (MAA, Level 1)

The next level in Martin Nunction, extending the array length of Martin Aunction, written as

$$A(a,b,c,d,e,\#)$$

(Note: octothorpe symbol means unchanged remainder of an array and hyperoperation.)

Where array entry must be a non-zero finite positive integer or an array.

General 4 Entry Expansion:

$$A(ab,c,d) = A(a,b,(a,b-1,c,d),d-1)$$

General Pairing rule for Entry >3 where there's a number 1 followed by a number 2, and where 1 followed by a number > 2, scanning right to left:

$$A(a,1,2,\#) = A(a,3,1,\#)$$

$$A(a,1,c,\#) = A(a,(a,(\dots\#\dots),c-1,\#),c-1,\#)$$

Where b is now replaced with an entire array nested into b entry c times, subtracting c by 1 on the outer array and all inner arrays

General n entries:

$A(a,b,c,d,\#) \rightarrow A(a,b,c,1,\#)$, Entries after d are for applying pairing rule such that any entry except for a = 1.

Example:

$$A(3,2,2,2) = A(3,2,(3,1,2,2),1) \rightarrow A(3,2,(3,1,2,2)), \text{ remove trailing 1, according to rule 3.}$$

→ Applying pairing rule, for pair $b=1$ and $c=2$, $(3,1)$.

$$A(3,2,(3,3,1,2))$$

→ Applying pairing rule, for pair $c=1$ and $d=2$, $(3,1)$.

$$A(3,2,(3,3,3,1)) \rightarrow A(3,2,(3,3,3)), \text{ remove trailing 1.}$$

→ Let $X = A(3,3,3)$

$$\rightarrow A(3,2,X) = 3 (X)^2 3.$$

3.Martin Array Nunction (MAN)

Martin Array Nunction(MAN)

$L_a(n, l)$.

let a = array level

let n = value of entries in current array level

let l = length or amount of entries in current array level

$L_1(n, 4) = A(n, n, n, n)$

$L_2(n, 3) = (n, n, n)$

general l = 1 leveling down: $L_a(n, 1) = L_{a-1}(n, n)$

examples:

$L_2(4, 1) = \backslash 4 \backslash = L_1(4, 4) = A(4, 4, 4, 4)$.

General l = 3 and 2:

$L_n(n, 2) = L_{n-1}(n, (L_{n-1}(n, n)))$

$L_n(n, 3) = L_{n-1}(n, (L_{n-1}(n, (L_{n-1}(n, n))))))$

$L_2(3, 2) = \backslash 3, 3 \backslash = L_1(3, (L_1(3, 3))) = \text{Let } X = L_1(3, 3), \text{ which is } A(3, 3, 3)$.

$\rightarrow L_1(3, X) = A(3, 3, 3, 3\#)$

General l = 4 expansion:

$L_n(n, 4) = (n, n, n, n) = (n, n, (n, n-1, n, n), n-1)$

General n length:

$L_n(n, n) = (n, n, n, n, \#) \rightarrow (n, n, n, 1, \#)$, more entries are for applying the Pairing rule.

4.Bangs (!)

An Array operator of MAN, the first priority Operator, written in the operator slots which are second to last entry in MAN.

$L_a(n,l,operator)$ or $L_a(n,l,!)$

General entry ≤ 3 with single or more bangs:

General Linear MAN array with single bang:

$(a,! ,b) = (a,a,...\#)$ or $L_a(a,2,!) = L_a(a,(L_a(b,b)))$

MAN current array filled with a with $L_a(b,b)$ Entries.

$(a,b,! ,c) = (a,a,...\#)$ or $L_a(a,3,!) = L_a(a,(L_a(b,(L_a(c,c))))$

MAN current array filled with a with $L_n(b,(L_n(c,c)))$ entries.

General MAN array with n bangs:

$(a,! ^{(n)},b) = (a,a,...\#,...,! ^{(n-1)},a)$ or $L_a(a,2,! ^n) = L_a(a,(L_a(b,b),! ^{n-1}))$

MAN current array filled with a with $L_n(b,b)$ Entries.

$(a,b,! ^{(n)},c) = a,a,...\#,...,! ^{(n-1)},k\backslash$ or $L_a(a,3,! ^n) =$

$L_a(a,(L_a(b,(L_a(c,c))),! ^{n-1}))$

MAN current array with $(n-1)$ bangs and $L_a(b,(L_a(c,c)))$ entries.

>3 entry with bangs recursion:

$(a,b,c,! ,d\backslash = \text{Let } X = \backslash a,b,! ,c\backslash, \text{ calculate 3 Entry with single bang first.}$

$\rightarrow \backslash X,X,X,! ,d-1\backslash$ or $L_a(a,4,!) = \text{let } X = L_a(a,3,!) \rightarrow L_a(X,3,!)$

General:

$L_a(n,l,!) = \text{Let } X = L_a(n,l-1,!) \rightarrow L_a(X,l-1,!)$

5. Tiers (/n/)

The second priority operator.

$L_a(n,l,/n/)$.

General Tiering:

$/n/ = \text{Tier 1 with value } n, //n// = \text{Tier 2 with value } n, (k\text{-th tier}(n) = k\text{-th tier with value } n, \text{ higher tier} = \text{less priority.}$

General tier 1 with $n = 1$ and $n > 2$:

$L_a(n,l,/1/) = \text{Let } X = L_a(n,l), \rightarrow L_a(n,l,! ^{(L_a(\#))})$

Evaluate X, then subtract tier 1 by 1, which becomes X amounts of Bangs, with entire function nested into it's exponent, where innermost nested function = $L_a(n, l, !^X)$

$$L_a(n, l, /n/) = \text{Let } X = L_a(n, l), \rightarrow L_a(n, l, /n-1/, !^{L_a(n, l)})$$

Evaluate X, then subtract tier 1 by 1, which becomes X amounts of Bangs added to the second rightmost entry as higher priority operator, with entire function nested into it's exponent, Where the innermost nested function = $L_a(n, l, /n-1/, !^X)$

General Tier 2 and Tier k with value n:

$$L_a(n, l, //n//) = \text{let } X = L_a(n, l), \rightarrow L_a(n, l, //n-1//, /(L_a(n, l, //n-1//, /X/))$$

$$\text{Where innermost nested function} = L_a(n, l, //n-1//, /X/)$$

Nest entire function X times inside Tier 1 bracket and subtract Tier 2 value by 1, then add lower tier operator on the right side of the higher tier in the operator slots

$$L_a(n, l, (k\text{-tier}(n))) = \text{let } X = L_a(n, l), \rightarrow L_a(n, l, (k\text{-tier}(n-1)), (k-1 \text{ tier}((L_a(n, l, (k\text{-tier}(n-1)), (k-1 \text{ tier}(X)))))$$

$$\text{Where innermost nested function} = L_a(n, l, (k\text{-tier}(n-1)), (k-1 \text{ tier}(X)))$$

Nest entire function into (k-1)-th tier bracket and subtract k-th tier value by 1, then add lower tier operator on the right side of the higher tier in the operator slots.

6. Martin Kunction (MK)

Index of Structure and tiers, written as:

$$M_k(m, n, l)$$

Let k = k-th tier structure within Martin Nunction.

Let m = tier within Current tier structure

Let n = amount of current tier within the structure

$$\text{Let } l = L_l(l, l)$$

$$M_k(m, n, 3) = (3, 3, (k\text{-th structure tier } m(n)), 3)$$

Example:

$$M_1(1, 1, 3) = (3, 3, /1/, 3) \text{ or } L_3(3, 3, /1/)$$

$$M_1(2, 1, 3) = \backslash 3, 3, //1//, 3 \backslash \text{ or } L_3(3, 3, //1//)$$

$$M_2(1, 2, 2) = (2, \{2\}, 2) \text{ or } L_2(2, 2, \{2\})$$

$M_2(2,2,2) = (2, \{\{2\}\}, 2)$ or $L_2(2,2, \{\{2\}\})$

$M_3(1,2,3) = (3,3,[2],3)$ or $L_3(3,3,[2])$

General Martin Kunction:

$M_k(m,n,l) = L_l(l,l,(k\text{-structure tier } m(n))) = \text{Let } X = L_l(l,l)$

$\rightarrow L_l(l,l,(k\text{-structure tier } m(n-1),(k\text{-structure tier } m-1(L_l(\#))))$

Calculate X, subtract k-th structure m-th tier value by 1, then add k-th Structure (m-1)-th tier with value X, nest entire function into it's bracket,

Where Innermost nested functikn = $L_l(l,l,(k\text{-structure } m(n-1)), (k\text{-structure tier } m-1(X)))$.

Leveling Down Structure:

$L_l(l,l,(k\text{-structure tier } 1(1))) = \text{Let } X = L_l(l,l)$

$\rightarrow L_l(l,l,(k-1 \text{ structure tier } X(L_l(\#))))$

Calculate X, Level down to (k-1)-th structure with tier X and Value X, nest the entire function into the (k-1)-th structure X-th tier bracket X times, with Innermost nested function = $L_l(l,l,(k-1 \text{ Structure tier } X(X)))$

7.Martin K Nunction (MKN)

Self-reference of MK, written as:

$M^w\{w\}_k(m,n,l)$

Let w = Amount of nesting parameter k.

$M^{w1}_k(m,n) = M_l(l,l,(M_k(m,n,l)\text{-th structure tier } m(n)))$,

$M^{w2}_k(m,n) = M^{w1}_l(M^{w1}_k(m,n))(m,n)$

General = $M^{w}_k(m,n) = M^{(w-1)}_l(M^{(w-1)}_k(m,n,l)(\#)(m,n))(m,n)$

Subtract w by 1, nest $M^{w-1}_k(m,n,l)$ w times as MK Exponent or w parameter.

B.Rules

Rule 1: MAN array entry must contain an array must contain non zero positive integers except for operator slot/value and MAN.

Operators Slot and Value or amounts.

Rule 2: $(\#, \text{Operators}, k)$ or $L_a(n,l, \text{operators}), b \backslash$ or $(a, [n], \{n\}, /n/, !, b)$ or $L_a(n,l, \{n\}, /n/, !)$.

Higher priority operator is written right before last entry, least Priority written after Second last array entry and so on, operator slots are always second to rightmost entry in MAN.

Rule 3: remove trailing 1, $(\#, 1) \rightarrow (\#)$.

Rule 4: Prioritize (!) before m-th tier, then k-th structure.

Rule 5: Evaluate Innermost array/nested function first.

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